

HRVOJE VIŠIĆ

CONTACT

Tübingen, Germany

ORCID 0000-0003-0406-5330

SKILLS

- R1 - Open water diver (SSI)
- Speleological caving course
- Drivers licence - B category
- Croatian (native), English (C1) German (A2), Italian (A1)

SOFTWARE

- ArcGIS, Grapher, Google Earth
- Adobe Creative Cloud, ImageJ
- QIIME, Mothur, BLAST, MEGA, Clustal, BioEdit, SnapGene
- RStudio, Statistica, Primer6
- Basic Python, R, JavaScript, HTML and YAML in VS Code
- NAS and Docker containers
- LLM prompt engineering
- Github, Oracle, SQL
- Match!, CrystalSleuth

CURRENT POSITION

SEP, 2022 - PRESENT

PhD Researcher - Geomicrobiology Group - DFG Emmy Noether Programme

My research investigates how microbial biosignatures from sulfur- and iron-cycling bacterial mats are fossilized through experimental taphonomy under hydrothermal conditions. I performed geochemical, microbiological, and mineralogical analyses to understand diagenetic processes relevant to ancient Earth and astrobiology. Additional responsibilities included IT support and management of website/server databases.

Doctorate at Department of Geosciences, University of Tübingen, Germany
Thesis: Experimental taphonomy of morphological biosignatures from sulfur and iron cycling bacterial mats under hydrothermal conditions
Supervisor: Dr. Muammar Mansor, Prof. Dr. Jan-Peter Duda, Prof. Dr. Andreas Kappler

WORK EXPERIENCE

JAN, 2022 – JUL, 2022

Research scientist
Geomicrobiology group, University of Tuebingen, Germany

- Biogenic iron sulfide formation by S-cycling microbes and microbial mats

SEP, 2019 – DEC, 2021

Scientific and professional associate
NovaDerm – NeoStratum, Zagreb, Croatia

- Dermatological R&D, quality control and database management

JUL, 2018 – AUG, 2019

Research and teaching scientist
Laboratory for Biological Oceanography, University of Zagreb - Croatia

- Fieldwork, eDNA metabarcoding, website, teaching and lab courses

TECHNIQUES

Field work:
- In situ physicochemical measurements
- Sediment coring
- Plankton-net filtration
- Speleological and bathymetric mapping

Laboratory skills:
- Glovebox and laminar flow work
- Sterile and anoxic solution preparation
- Radiation protection training

Mineralogy & geochemistry
- Sequential extraction/elemental speciation
- Synthesis of colloids and iron minerals
- Sulfate turbidimetry, silicomolybdate, sulfide and ferrozine assay via UV–Vis
- µRaman, µXRD, FTIR, ICP-MS, SIMS
- Micropetrographic thin-section analysis
- Lipid biomarker extraction via LC column
- Zeta potential and magnetic susceptibility

Paleoecology:
- Paleoenvironment reconstruction from biostratigraphy and lithological succession

Microscopy:
- Light, polarization and epifluorescence
- SEM–EDX for minerals and microbes

Microbiology:
- Enrichment and pure culture isolation
- Cloning and transformation
- CARD-FISH and flow cytometry
- Synchrotron-based Cryo 3D SIM-SXT

Molecular biology:
- eDNA isolation and metabarcoding
- PCR and gel electrophoresis
- Sequence data analysis (16S, 18S, ITS)
- Western blot and protein purification

Ecotoxicology:
- Genotoxicity comet test
- Zebrafish embryotoxicity test

EDUCATION

SEP, 2015 - FEB, 2018

Master in Environmental sciences
Faculty of Science, University of Zagreb

Thesis: Chemotaxonomic and morphological approach to phytoplankton analyses in contrasting trophic systems of North Pacific
Supervisor: Prof. Dr. Zrinka Ljubešić

SEP, 2012 - SEP, 2015

Bachelor in Environmental sciences
Faculty of Science, University of Zagreb

Thesis: Geological evidences of previous habitability on the surface of Mars
Supervisor: Prof. Dr. Dražen Balen

INTERNSHIPS

JUL – SEP, 2016

Erasmus+ traineeship at DNA Repair Mechanisms and Human Disease Group
- University of Oxford
- Supervisor Dr. Ivan Ahel

2015 – 2016

Laboratory internship at Evolution, Symbioses and Molecular Phylogenetics
- University of Zagreb
- Franjević et al. 2019.

2014 – 2015

Laboratory internship at Ecotoxicology Group
- University of Zagreb
- Babić et al. 2017.

2013 – 2018

Student rights representative on the Biology Department Council
- University of Zagreb

TEACHING

University courses
- Alpine geomicrobiology field course
- Geomicrobiology laboratory course
- Microbiology of the ecosystem
- Pelagial microbiology
- Microbial ecology
- Invertebrates (TA)
- Protists (TA)

Scientific outreach
- Lab science for school kids 2024
- Day and Night at PMF 2019
- Night of Biology 2014–2017
- Open days of IRB 2015
- Science Picnic 2015

Bioteka Association 2020–2021
- NGO for promotion of sciences for talented children - Zagreb, Croatia

BIUS Astrobiology Section mentor

WORKSHOPS

JUN, 2025

Biominerals workshop
- University of Tuebingen

JUL, 2023

Tools in Biogeochemistry
- University of Tuebingen

MAR, 2023

Biosignatures Workshop
- University of Tuebingen

JAN, 2023

SwissSIMS Winterschool
- University of Lausanne

MAR, 2020

Precambrian Paleoceanography
- University of Zagreb

MAR, 2019

Eukaryotic Metabarcoding
- Physalia courses, Berlin, Germany

FEB, 2019

DNAqua-Net Diatom Biomonitoring
- INRAE, Thonon-les-Bains, France

NOV, 2018

University Computing Centre Zagreb
- Introduction to Linux
- Computational cluster Isabella
- Programming basics in Python & R

NOV, 2015

2nd International Conference on Evolution and Behaviour
- Part of technical team
- University of Zagreb

NOV, 2015

International Student Conference on Environmental Sciences
- Part of organizing team
- University of Zagreb

FIELDWORKS

APR, MAY 2023, FEB 2024

Abandoned mines sampling
- Hoffnungsstollen and Segen Gottes, BW, Germany

AUG, 2022, JUN, SEP 2023

Alpine streams sampling
- Engadin, Switzerland

JUN, 2017

Lost Lake Landscapes of the Eastern Adriatic Shelf (LoLADRIA)
- Telašćica Nature Park, Croatia

MAY, 2016

Education on the developmental potentials of the Croatian islands
- The island of Cres, Croatia

JAN 2016

Study of the endemic species of *Marifugia cavatica* and *Eunapius subterraneus* in the Tounj cave system and ecological conditions

AUG, 2015

Scientific-educational research camp Papuk Nature Park, Croatia
- Biology Students Association

MAY, 2014

Dinaric Glaciation 2nd workshop Paklenica National Park, Croatia

SCIENTIFIC PUBLICATIONS

Rector's award for individual scientific and artistic work (University of Zagreb)

- **Višić, H.** 2017. Toxicity of contaminated sediment from the Sava River on zebrafish embryos (*Danio rerio*)

Peer-reviewed journal articles

- **Višić, H.**, Mansor, M., Fischer, S., Kappler, A., Duda, J.-P. Survivability of experimentally fossilized filamentous microbial morphologies under low metamorphic conditions using HPHT autoclave - *in prep*.
- Sekerci, F., Wilsch-Bräuninger, W., **Višić, H.**, Fischer, S., Peiffer, S., Kappler, A., Mansor, M. Biogenic pyrite formation under archean-like ferruginous conditions. - *in prep*.
- Mansor, M., **Višić, H.**, Dreher, C.L., Zirner, A., Markl, G., Mentzer, S.M., Duda, J.P., Escudero, C., Kappler, A. Geomicrobiology of iron-rich stalactites: Implications for habitability of iron-rich deposits on Mars. - *Submitted*.
- **Višić, H.**, Mansor, M., Fischer, S., Sekerci, F., Ilin, A. M., Little, C.T.S., Kappler, A., Duda, J.-P. Iron-oxidizing bacteria as overlooked sources of pyrite microfossils: Evidence from experimental diagenesis of tubular sheaths and twisted stalks. - *Submitted*
- **Višić, H.**, Duda, J.-P., Fischer, S., Escudero, C., Sekerci, F., Kappler, A., Mansor, M., 2025. Experimental diagenesis reveals preservation of biosignatures in filamentous sulfur mats under hydrothermal conditions. *Sci Rep 15*, 38114.
- Matek, A., Bosak, S., Šupraha, L., Neeley, A., **Višić, H.**, Cetinić, I. and Ljubešić, Z., 2023. Phytoplankton diversity and chemotaxonomy in contrasting North Pacific ecosystems. *PeerJ*, *11*.
- Franjević, M., Hrašovec, B., Blašković, P., **Višić, H.**, Franjević, D., 2019. Nova metoda genske identifikacije vrsta u integriranoj zaštiti šuma. *Naše Šume, 17(54/55)*, 45-48.
- Babić, S., Barišić, J., **Višić, H.**, Klobučar, R.S., Popović, N.T., Strunjak-Perović, I., Čož- Rakovac, R., Klobučar, G., 2017. Embryotoxic and genotoxic effects of sewage effluents in zebrafish embryo using multiple endpoint testing. *Water research, 115*, 9-21.

Oral presentations at conferences

- **Višić, H.**, Fischer, S., Duda, J.P., Kappler, A. and Mansor, M., 2025. From the depths of the Earth to beyond: preservation potential of morphological iron biosignatures under diagenetic conditions. *Biennial European Astrobiology Conference (BEACON).*
- **Višić, H.**, Duda, J.P., Kappler, A. and Mansor, M., 2024. Forensic Astrobiology–Experimentally Assessing the Preservation of Microorganisms Encompassing Hydrothermal Vent Systems. *Astrobiology Science Conference (AbSciCon)* (pp. 103-06).
- **Višić, H.**, 2020. How to watch the plankton from space. *In 23rd Symposium of Biology Students in Europe book of abstracts*, 54. Groningen, Netherlands (virtual).
- **Višić, H.**, Filek, K., Mucko, M., Trotta, A., Panagopoulou, A., Lukač, M., Corrente, M., Di Bello, A., Gračan, R. and Bosak, S., 2019. Metagenomic characterization of the surface biofilm on mediterranean loggerhead sea turtles. *In Seventh European Phycological Congress Programme and Book of Abstracts*, 59. Zagreb, Croatia (presentation).
- **Višić, H.**, Babić, S., Barišić, J., Klobučar, G., 2017. Impact of wastewater effluents and contaminated sediments on zebrafish embryos using multiple endpoint testing. *In 20th Symposium of Biology Students in Europe book of abstracts*, 52–53. Lund, Sweden.

Proceedings of abstracts at conferences

- Mansor, M., **Višić, H.**, Runge, E., Shuster, J., Fischer, S., Kappler, A. and Duda, J.P., 2023, October. Microbial pyrite formation: mineral morphology and precipitation kinetics. *In ARPHA Conference Abstracts* (Vol. 6, p. e106753).
- Runge, E., Mansor, M., Kappler,A. , Pei, Y., **Višić, H.**, Domingos, J.-M., Werner, P. and Duda, J.-P., 2022: The taphonomic fate of biominerals in hydrothermal sulfide systems – implications for the reconstruction of microbial life in deep time. *In GeoMinKöln 2022*, Cologne, Germany (presentation).
- **Višić, H.** and Mansor, M., 2022: Minerals and life - using pyrite as an indicator of microorganisms. *In SPP1833 Building a habitable Earth Abstract of the Sixth General Meeting "Early Earth evolution",* pp. 21., Tuebingen, Germany (poster).
- Van de Vijver, B., Filek, K., Gračan, R., Mucko, M., **Višić, H.**, Robert, K., Majewska, R., Frankovich, T.A., Ashworth, M.P., Bosak, S., 2019. The TurtleBIOME project: insight into epizoic diatom communities on loggerhead sea turtles. *In 24th Nordic Diatomist Meeting-Abstracts*, pp. 11., Stockholm, Sweden (presentation).
- Bosak, S., **Višić, H.**, Filek, K., Robert, K., Van de Vijver, B., Trotta, A., Panagopoulou, A., Majewska, R. 2019. DNA metabarcoding and morphological analyses of the diatom biofilm associated with loggerhead sea turtles in the Mediterranean Sea. *In The Molecular Life of Diatoms-Programme and Abstract Book*, pp. 104., Norwich, United Kingdom (poster).
- Bosak, S., Gračan, R., Mucko, M., **Višić, H.**, Filek, K., Gobić Medica, K., Mičić, M., Lukač, M., Basu, S., Orlić, S., Majewska, R., Frankovich, T. A., Ashworth, M. P., Van de Vijver, B., 2018. Loggerhead sea turtle microbiome – TURTLEBIOME project: insight into endozoic and epizoic communities. *In 6th Mediterranean Conference on Marine Turtles*, pp. 152., Poreč, Croatia, (poster).
- **Višić, H.**, Ljubešić, Z., Bosak, S. Neeley, A. R., Durkin, C. A., Estapa, M. L., Omand, M. M., Mucko, M., Barešić, A., Karwowska, Z., Cetinić, I., 2018. Multilayer approach to plankton analyses in contrasting trophic systems of North Pacific. *In Ocean Optics XXIV, 249*, pp. 1., Dubrovnik, Croatia, (poster).
- Ljubešić, Z., Bosak, S., Šupraha, L., Neeley, A.R., Durkin, C., Estapa, M., Omand, M., Mejdandžić, M., Barešić, A., Karwowska, Z., Cetinić, I., **Višić, H.**, 2018. In Depth Analyses of Phytoplankton Community in Northern Pacific During the Sea2Space Particle Investigation. *In Ocean Sciences Meeting*, Portland, Oregon, USA (poster).
- Babić, S., Sauerborn Klobučar, R., Barišić, J., Klobučar, G., Kepec, S., **Višić, H.**, Barišić, J., Topić Popović N., Strunjak-Perović I., Čož-Rakovac, R. 2015. Evaluation of toxic potential of municipal-waste waters from wastewater treatment plant by zebrafish embryo toxicity test. *In 12th Croatian Biological Congress book of abstracts*. Sveti Martin na Muri, Croatia, pp. 247-248 (poster).